# Grant Long

PhD Student,
Stanford University, CA,
207-209-8152, glong1@stanford.edu


## Professional Summary

I am a dedicated geologist with a keen focus on the dynamics of landscapes in collisional margins and their role in shaping landscape evolution over geologic timescales. My primary research interests lie at the intersection of climate and tectonics, where I explore how these forces drive change in Earth's topography. My research approach is highly field-oriented, involving intensive field seasons to gather samples, followed by comprehensive analysis at Stanford facilities. I am skilled in hands-on data collection and interpretation, with a focus on translating results into meaningful geological insights.

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## Education

**Ph.D. in Earth and Planetary Sciences, *in progress*** **Fall 2024 - Present**
Stanford University, CA,
Department of Earth and Planetary Sciences

**M.S. Geological Sciences** **Completed 2024**
Stanford University, CA,
Department of Earth and Planetary Sciences

**B.S. Geology** **Completed 2022**
University of Vermont, VT,
Department of Geology

## Research

My current research involves interpreting erosion dynamics in south-central Chile by combining the use of in-situ cosmogenic radionuclides, beryllium-10 specifically, to measure erosion rates within the Maule Region and the use of detrital zircon geochronology to understand the provenance. I present a novel methodologic approach to understanding the fertility of both quartz and zircon within a watershed to better understand and characterize their distributions. My research aligns with certain goals of the SZ4D initiative within the Landscapes and Seascapes working group to understand landscape evolution better by seeking answers from the past.

## Ongoing Projects

**Overcoming a topographic divide; rivers** : Modeling of how a river can overcome a topographic divide such as the Coastal Cordillera in Chile, where the rivers transverse the range, in comparison to the Willamette Valley in Oregon where the Willamette River runs parallel to the coast range.

**Base level change on the Rapel River** : Through the use of cosmogenic radionuclides I am looking at the changes in the Rapel River in response to recent uplift where there is evidence in the landscape from paleo relief, pediplanation, and modern incision into the pediplain.

**Erosion rate distributions** : Summer of 2024 a large collection of modern river samples was taken from the multiple rivers in the Maule Region of Chile to understand the erosion rate patterns from the Principal Cordillera to the coast and the contribution to the mass balance of this portion of the subduction zone.

**Sediment mixture modeling** : Using 10-Be and detrital zircon to measure watershed fertility and erosion dynamics within the Mataquito River watershed in south-central Chile. Future goals include understanding the distribution of watersheds and their sources from less than 100,000 years ago and how they have developed.

**Impacts of the mass balance in convergent margins, case studies from the Andes** : Understanding the unique tectonic history and fluctuations in orogenic mass balance, especially concerning magmatic flux and retroarc accretion, that is likely the dominant factors shaping the Andes, and perhaps other convergent margins as well.

## Teaching Experience

**Teaching Assistant** **Summer 2025**
*Stanford University, Department of Earth and Planetary Sciences*

Course: Introduction to field methods. Field-based lessons and discussions. I lead different parts of the trips and teach field-based skills and techniques.

**Trip Leader** **Winter 2025**
*Chile SZ4D Field Trip*

Lead the final day of the Chile SZ4D field trip where I brought participants through the "sink" of the Mataquito River looking at gravel deposits and sedimentary sequences on the coast.

**Teaching Assistant** **Fall 2024**
*Stanford University, Department of Earth and Planetary Sciences*

Course: Moving and Shaking in the Bay Area. Field-based lessons and discussions. I lead different parts of the trips and teach field-based skills and techniques.

**Travel Sophomore College Assistant** **Summer 2024**
*Stanford University Sophomore College*

Course: Mining and the Green Economy. Led laboratory discussions and aided in field-based trips while in Montana.

**Teaching Assistant** **Winter 2024**
*Stanford University, Department of Earth and Planetary Sciences*

Course: Structural Geology. Graded and provided feedback on assignments regarding structural geology topics. Held weekly office hours to support students.

**Teaching Assistant** **Fall 2022**

*Stanford University Department of Earth and Planetary Sciences*

Course: Moving and Shaking in the Bay Area. Field-based lessons and discussions. Led different parts of the trips and taught field-based skills and techniques.

## Conference Participation

Long, G., Aron, F., & Hilley, G. E. (2025). Erosion dynamics and sediment distribution on the Mataquito River, Chile. American Geophysical Union Conference, New Orleans LA. 2025. (talk)

Long, G., Aron, F., Hilley, G. E. (2024). Erosion Dynamics in South-Central Chile Using Detrital Zircon and Quartz. American Geophysical Union Conference, Washington DC. (poster)

Long, G., Aron, F., Willenbring, J., Hilley, G. E. (2023). Investigation of Erosion Dynamics in the Chilean Andes using Cosmogenic Radionuclide Inventories and Detrital Zircon U-Pb Geochronology. American Geophysical Union Conference, San Francisco, CA. (poster)

Hilley, G. E., Aron, F., Baden, C. W., Dobbs, S. C., Long, G. (2024). Impacts of the Mass Balance in Convergent Margins on Coupling Between Surface Processes and Tectonics, with Case Studies from the Central Andes. American Geophysical Union Conference, San Francisco, CA. (poster)

Hilley, G. E., Long, G., & Aron, F. (2025). Active Coastal Uplift in South-Central Chile Imaged by Erosion Rates and River Profiles in the Rapel River Watershed. American Geophysical Union Conference, New Orleans LA. (poster)

## University Service

**Stanford University**

Travel Sophomore College Assistant — Summer 2024
Course: Mining and The Green Economy

Stanford Doerr School of Sustainability Recruitment Ambassador — Fall 2023 to present
Earth and Planetary Sciences Department

Stanford Earth Explorers Club President — Fall 2022 to present

## Skills

- Well-versed in the procedures of mineral separation.
- Fluent in Adobe Suite application products.
- Familiar with Python programming language and the applications to sediment mixture modeling.
- Familiar with QGIS software for creating maps and data sets.
- Strong field geology knowledge and practical skills both in teaching and execution.
- Interpersonal skills and ability to work in group settings.

## Personal interests

Growing up in southern Maine, I was always outdoors regardless of the weather because I'd rather be outside and experiencing nature than inside. It led me to seek outdoor adventures such as skiing, hiking, and climbing. They are my favorite activities and have brought me to some spectacular locations. My

passion for travel is strong because I have an innate desire to learn about new places and people to further my knowledge of cultural differences and similarities. As a well-rounded person, I can relate to, enjoy, and embrace different perspectives and ideologies. These activities and passions have led me to the field of geology. Understanding how the landscapes I look at and experience change over time allows me to combine my passion for the outdoors with my knowledge of geology to create a lasting impact on the field of geology with the hopes that the research I do creates a better place for future generations to live through an understanding of why things work the way they do.